



Indian Institute of Information Technology, Surat

भारतीय सूचना प्रौद्योगिकी संस्थान, सुरत

(Institute of National Importance under Act of Parliament)

Department of Computer Science and Engineering
B.Tech- 4th Semester , End Semester Examination
Subject & Code: Software Engineering (CS-401)

Date: 5th May 2025

Time : 10.00 am - 1.00 p m

Marks: ____/100

Note: All questions are compulsory.

1. Assumptions can be made with proper justification.
2. Be precise and clear in answering the questions.

Questions

Marks

1.

a) What do you understand about software configuration management? How is it carried out?

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b) Assume that the size of an organic type software product has been estimated to be 32,000 lines of source code. Assume that the average salary of a software developer is Rs. 15,000 per month. Determine the effort required to develop the software product, the nominal development time, and the cost to develop the product.

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2.

a) Enumerate the different types of cohesion that a module in a design might exhibit. Give examples of each.

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b) In a shopping cart system, the "Checkout" module directly accesses the "Inventory" module's internal variables to update stock. What type of coupling is this? What problems can arise, and how would you redesign it?

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3.

Explain how the DFD model can help one understand the working of a software system. Draw a level 0 and level 1 DFD for a Food Delivery System where users browse restaurants, place orders, track delivery status and make payments online.

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4.

a) What are the activities carried out during testing a software? Schematically represent these activities. Which one of these activities takes the maximum effort?

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b) What is the difference between black-box testing and white-box testing? Give an example of a bug that is detected by the black-box test suite, but is not detected by the white-box test suite, and vice versa.

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5. List and explain three common types of risks that a typical software project might suffer from. Suppose you are the project manager of a large software development project, point out the main steps you would follow to manage risks in your software project.

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6. Write a C function to sort an array of 100 integers in ascending order using a sorting algorithm.

(i) Draw the Control Flow Graph (CFG) for your sorting function.

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(ii) Determine its Cyclomatic Complexity.

7. Distinguish between software verification and software validation. Can one be used in place of the other? Justify your answer. In which phase(s) of the iterative waterfall SDLC are the verification and validation activities performed?

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8. Describe the different phases of Agile Software Development Life Cycle (SDLC). Explain with an example how continuous feedback and iteration help in improving the final product.

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9. Explain the concept of software reliability. How does it differ from hardware reliability? Illustrate your answer with an appropriate diagram comparing the failure behavior of hardware and software systems over time.

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10. What is Six Sigma? Explain its significance with the help of a normal distribution curve. Describe the key characteristics that differentiate Six Sigma from traditional quality management approaches.

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